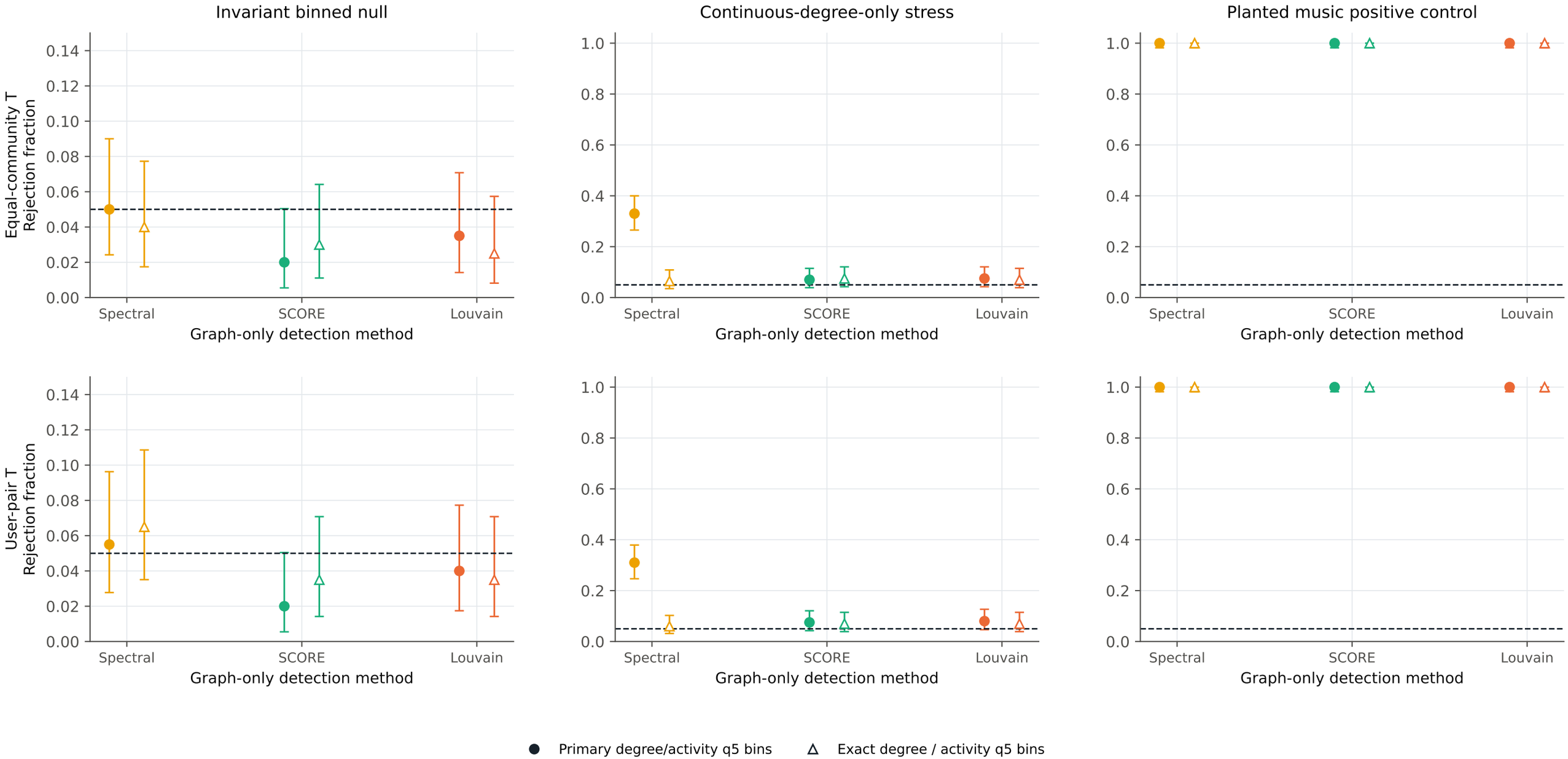
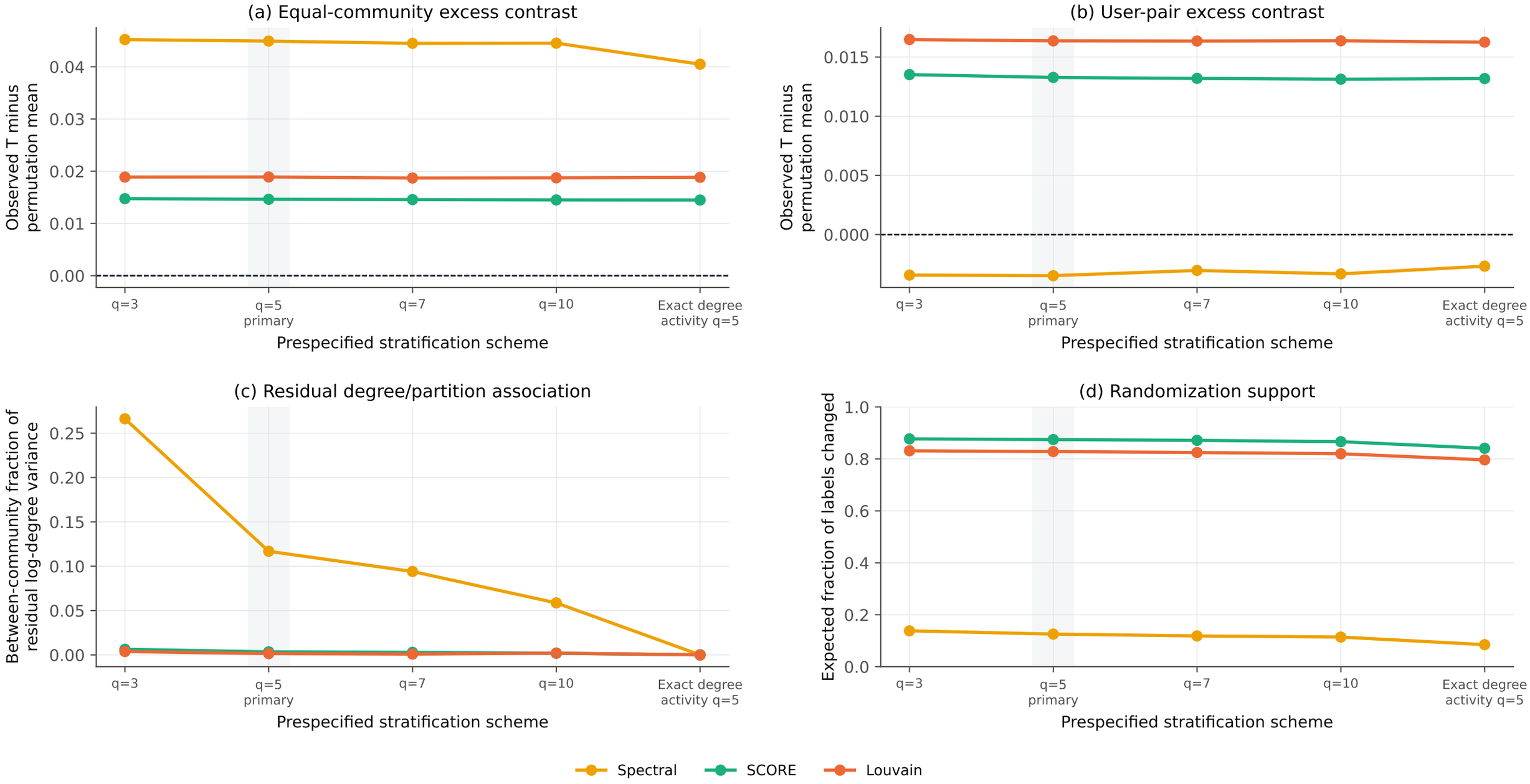
Does the primary permutation procedure behave as intended?



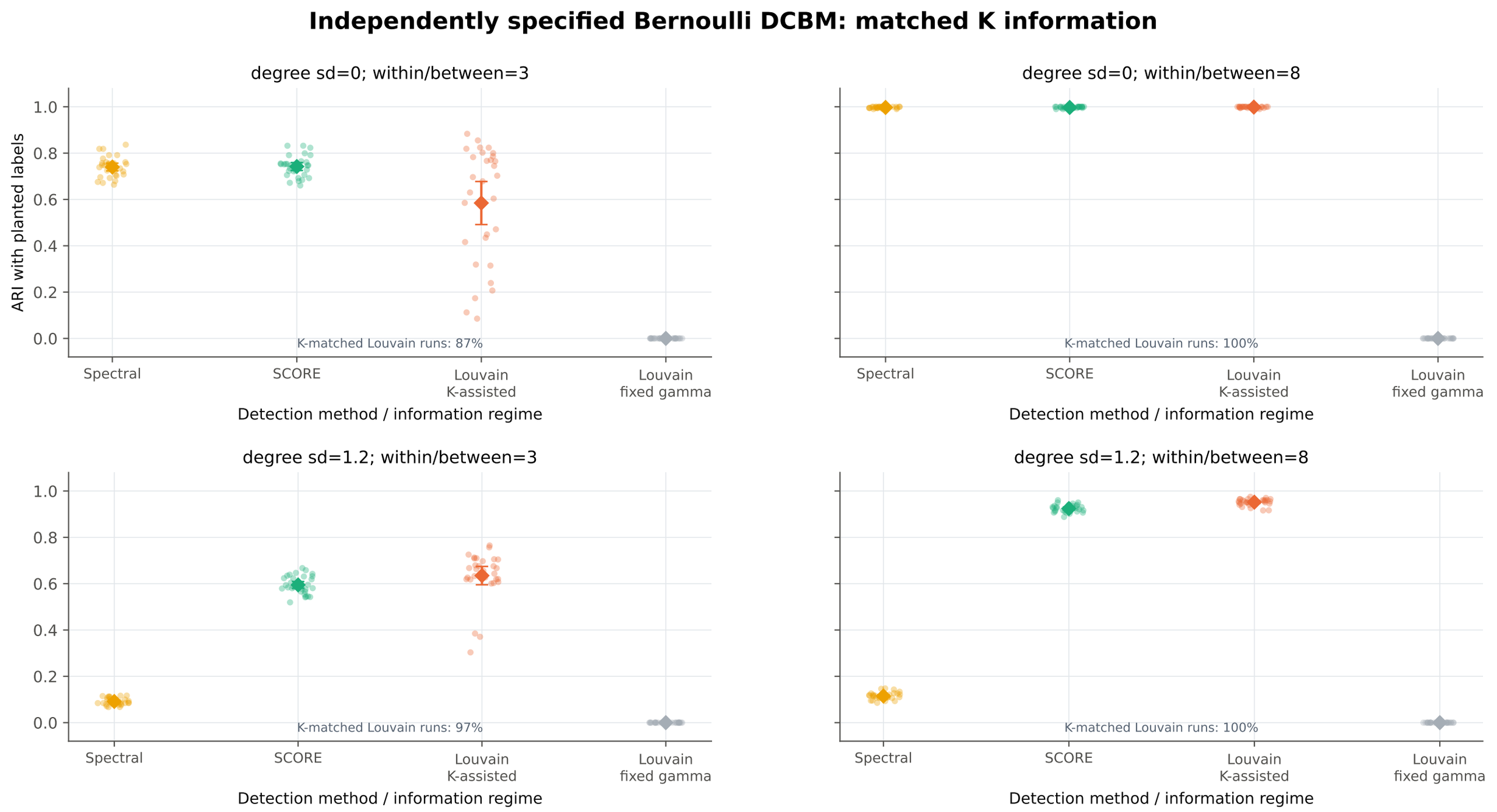
200 independent music replications; 199 permutations each; bars: pointwise binomial Monte Carlo 95% intervals. Dashed line: alpha=.05.
Middle column: no community term beyond exact degree; q5 is approximate, so its rejection rate is a stress diagnostic, not exact-null size.
Left: both references are invariant nulls. Right: rejection measures power. These experiments do not establish exchangeability in LastFM.

How sensitive is community/music inference to nuisance stratification?

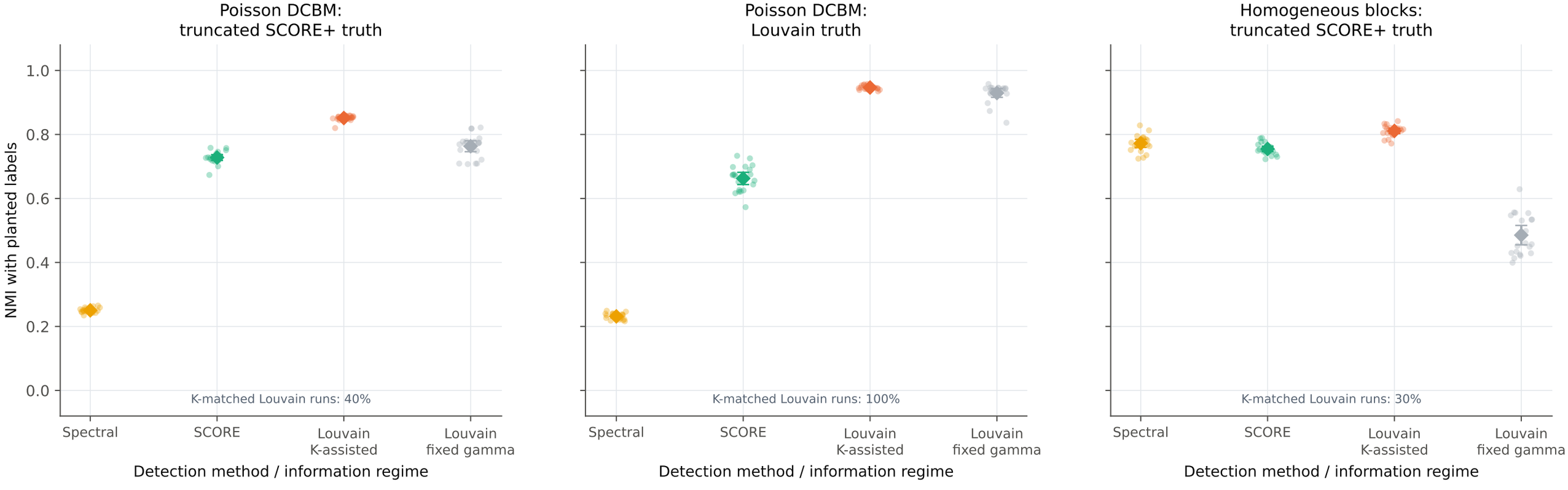


LastFM Asia, March 2020 snapshot. Primary q5: B=1999; other references: B=499.

Comparisons are exploratory. Exact degree does not mean exact activity adjustment. Residual covariate association is diagnostic, not a test of music exchangeability.

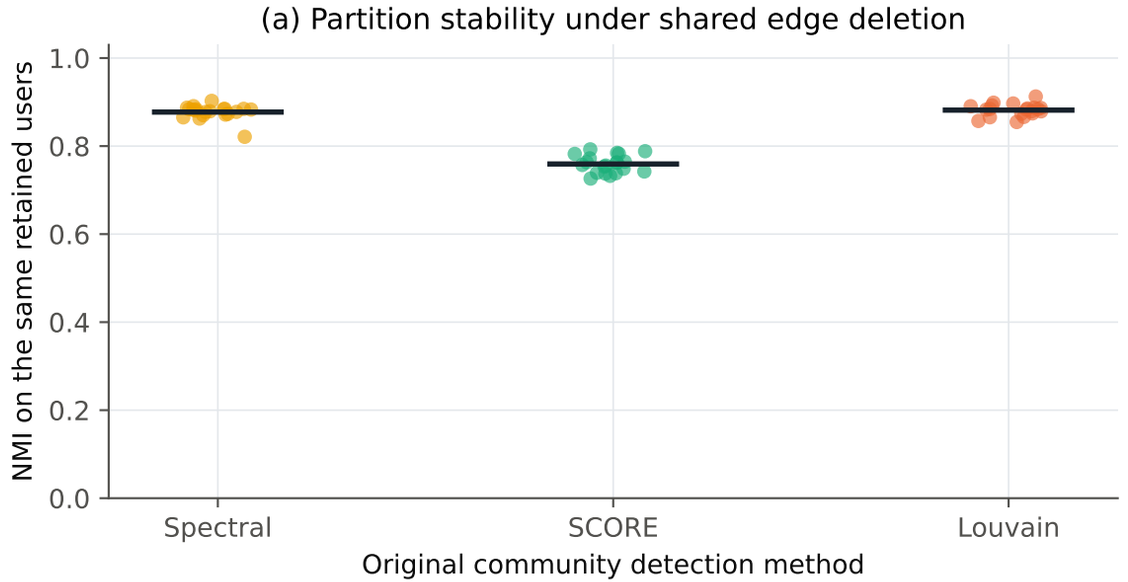

Independently specified Bernoulli DCBM: matched K information
degree sd=0; within/between=3
degree sd=0; within/between=8
degree sd=1.2; within/between=3
degree sd=1.2; within/between=8
ARI with planted labels
1.0
0.8
0.6
0.4
0.2
0.0
K-matched Louvain runs: 87%
K-matched Louvain runs: 100%
K-matched Louvain runs: 97%
K-matched Louvain runs: 100%
Spectral
SCORE
Louvain
K-assisted
Louvain
fixed gamma
Detection method / information regime
30 paired graph replications per setting; three planted groups; expected full-graph mean degree 18.
Same connected evaluation population across degree/signal regimes within each replication; probabilities are never clipped and graphs are not redrawn.
Diamonds/bars: mean and pointwise Monte Carlo 95% interval. Gray fixed-resolution Louvain is a reference, not part of the matched-information comparison.

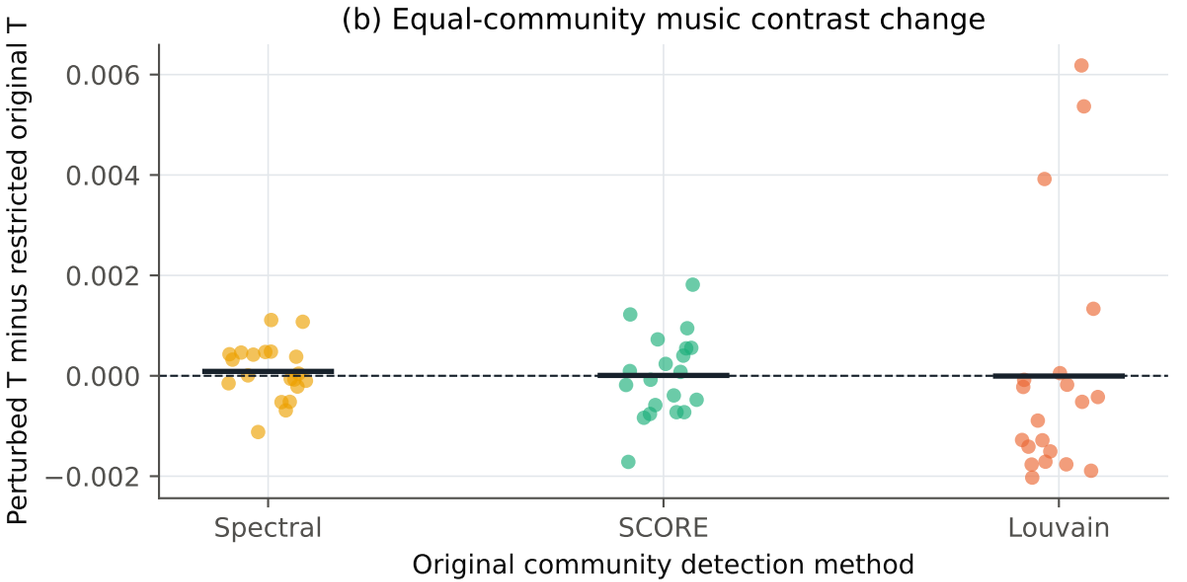
Does unequal K information change the calibrated recovery comparison?

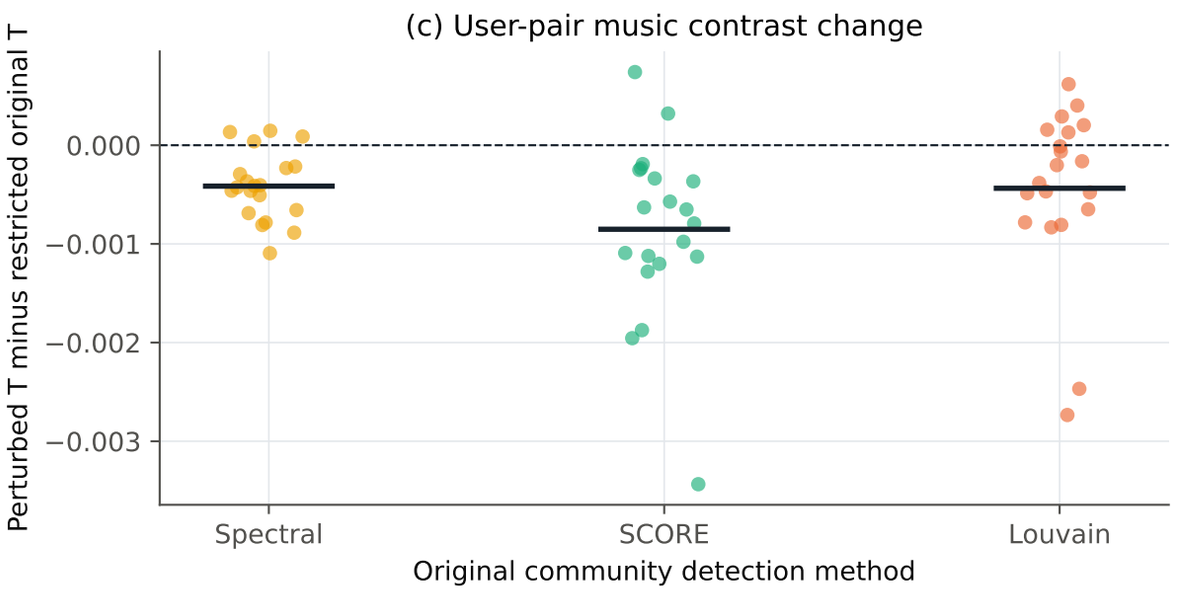


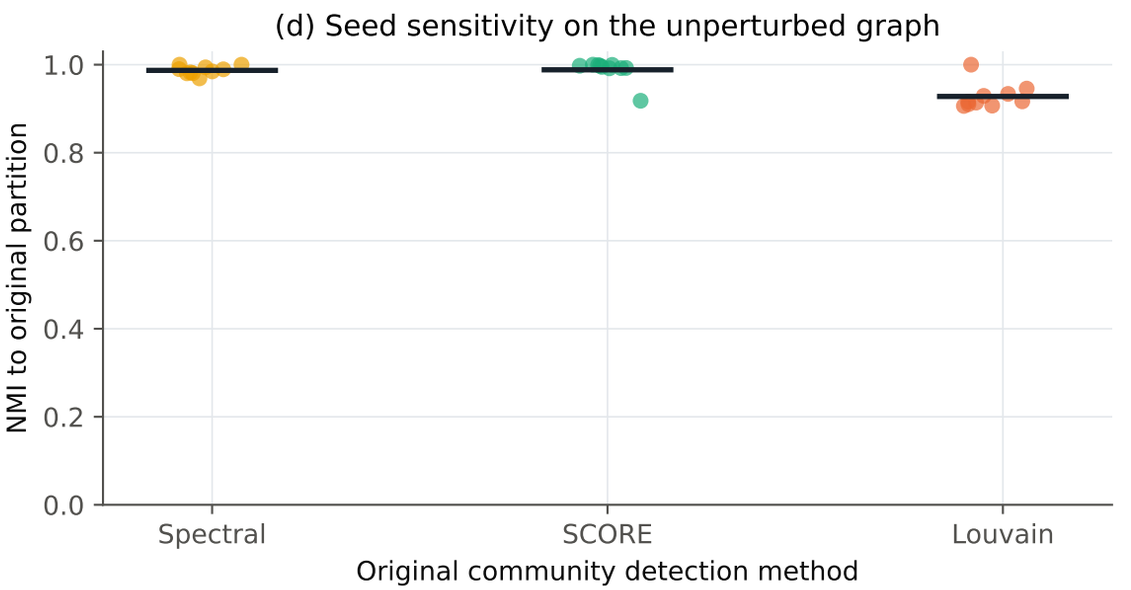
20 reproduced legacy graphs per setting; all methods evaluated on identical retained users in each graph.
Louvain K assistance selects resolution using graph cluster counts only, never membership truth. Actual K matches are recorded.
Points: individual graphs; diamonds/bars: mean and pointwise Monte Carlo 95% interval. These are generator-conditional recovery results.

Are the community assignments and music contrasts stable?









LastFM Asia, March 2020 snapshot: 20 common 10% edge-deletion masks; 10 random seeds per method.
Retained users: 96.7% to 97.4%. Each T change compares identical retained users.
Points are perturbations/seeds; black bars are means. These are algorithmic sensitivity diagnostics, not population confidence intervals.